

Impact of soil management systems on soil structure and physical properties in a clay loam soil, and the simulated effects on water deficits, soil aeration and workability

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ABSTRACT

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The effects of different soil management systems on soil structure and associated physical properties in a Dutch clay loam soil were evaluated using a combination of simulation modelling, field investigations and laboratory experiments. The investigations were focused on field recognition of identical soils only differing in structure type caused by soil management systems. Two soil structure types were recognised: a type developed in permanent grassland and a type in young arable land, which had been used for intensive vegetable growing since 1979. The types were characterised by examining the micro- and macro-morphology and soil hydraulic properties, such as soil structure, rooting patterns, pore shape and size distribution, bypass flow and water retention and unsaturated hydraulic conductivity characteristics. Field recognition of structure types and subsequent morphological and physical characterisation was used as an alternative to the application of a machine–soil submodel.

Significantly different properties of both soil structure types were functionally interpreted by using a simulation model for the soil–plant subsystem. Morphological and hydraulic analyses were essential for the definition of proper boundary conditions and input parameters. The land qualities water deficits, workability and soil aeration were calculated for a potato crop under Dutch climatological conditions. Expressions for temporal variability were obtained by using a 30 year record of weather data. The soil structure of permanent grassland proved to be more favourable than that of young arable land. This resulted in a 10% decrease in the average annual water deficit, 40% more workable days in the planting phase and 5–10% more days with a well-aerated soil. The impact of these differences on the potato tuber yield interpreted in terms of the effect of soil structure degradation, was estimated to be at least 5% in 20 out of 100 years under Dutch climatic conditions. The differences calculated also equalled the maximum possible effect of soil structure regeneration.

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INTRODUCTION

In the past two decades, root and field vegetable crops superseded cereals as the dominant crops in the Netherlands, detrimentally affecting the soil structure of Dutch arable land. This degradation of soil structure is chiefly the result of later harvesting, usually under wetter soil conditions, and the production and field transport of a larger quantity of harvestable products. Despite a considerable research effort, quantitative knowledge about interactions between mechanisation, soil conditions and crop growth, is as yet insufficient (e.g. Boone, 1988; Håkansson et al., 1988). Continuous research is needed to investigate the highly complex machine-soil-crop system. One way of expanding know-how would be by conducting field experiments. These experiments, however, are too expensive to cover the whole range of prevailing crops, soil and weather conditions and the different field traffic and tillage actions. Hence, a combination of field experiments and simulation modelling could be an attractive alternative to extrapolate the experimental results obtained at one site to a wider range of site conditions.

The current models that simulate the total machine-soil-crop system, have serious shortcomings (e.g. Gupta and Allmaras, 1987; Hadas et al., 1988). Accurate prediction of soil structure and quantification of soil pore geometry and associated physical properties, after using a tillage implement, is impossible. Although the comprehensive models are not yet operational, validated segments, such as the machine-soil submodel and the soil-plant submodel, may be applied. For example, field measurements of physical characteristics and the use of a soil-plant simulation model to investigate water-associated effects of different soil management systems have been put into practice by Van Lanen et al. (1987), Simota and Canarache (1988) and Van Lanen and Boersma (1988).

Our study set out to quantify the effects of different tillage and field traffic systems on land qualities, such as water deficit, workability and soil aeration for soil structure types in clay loam soils. Two soil structure types were considered, permanent grassland (G) and young arable land used for market gardening (A). These structure types occur in otherwise identical soils, being a result of different soil management systems. We investigated the effects of relatively recent changes in soil structure. Differences in simulated land qualities and crop yields provided quantitative and functional information on soil structure degradation caused by different soil management systems. Any positive effects of soil structure regeneration were also explored. More details on our study were provided by Reinds (1988a). Results for sandy loam soils under Dutch climatic conditions were reported by Van Lanen et al. (1987) and Van Lanen and Boersma (1988).

METHODS AND MATERIALS

Soils, soil structure types and physical properties

Clay loam soils cover about 190 000 ha in the Netherlands. The clay loam soils considered in our study were located near Piershil, about 15 km south of Rotterdam. The soils had developed in calcareous, young marine deposits and were classified as Typic Fluvaquents (Soil Survey Staff, 1975) and as Calcaric Fluvisols (FAO-UNESCO, 1974). The clay content of the topsoil was about 30% and decreased with depth to about 18% at 1 m below the soil surface. The organic matter content and the pH of the topsoil were 2.4% and 7.5, respectively. More analytical data have been provided by Reinds (1988a,b) and Van Lanen and Boersma (1988). The soils were well drained; the mean highest water table (MHW) was at a depth of 60–70 cm and the mean lowest water table (MLW) at 160–180 cm below the soil surface (Van der Sluijs and De Gruijter, 1985). In the Netherlands, soils such as these are mainly used to grow arable and field vegetable crops. Such small areas can be found in permanent grassland in the vicinity of farms.

Two different soil structure types in clay loam soils were considered in our study, a soil structure type G and soil structure type A. Until 1979 both soils were identical and had been used as permanent grassland for decades, when type A was changed from permanent grassland to land for market gardening. The experimental sites with the different soil structure types were adjacent plots, less than 100 m apart. In the period 1985–1988 field and laboratory experiments were conducted to characterise both soil structure types. The structural features of the two types are presented in Fig. 1. The structure description of type A holds for the situation in 1987, while the description for G applies to any time. During the experimental period, the perennial vegetable crop rhubarb was grown on young arable land, after 6 years of very intensive vegetable growing of usually more than one crop per year. The notable differences in soil structure of the topsoil were clearly visible in the field, as illustrated in Fig. 2. Soil structure was further characterised in the laboratory by the electro-optical image analyses, using thin sections (Kooistra et al., 1984, 1985). The pore geometry was described in terms of pore shape and size.

Because differences in pore morphology could affect physical properties of the soil, the water retention and unsaturated hydraulic conductivity curves were measured using undisturbed soil samples from each soil structure type. The samples were gathered at three depths: the topsoil (0–25 cm); an intermediate layer (25–45 cm); the subsoil (more than 55 cm). The samples were replicated five- or six-fold. The samples of the permanent grassland site had been collected in 1985. The young arable land was sampled in 1987 after the ploughed topsoil had sufficiently settled following planting of the rhubarb crop. Thus, morphological and physical properties of young arable land apply to

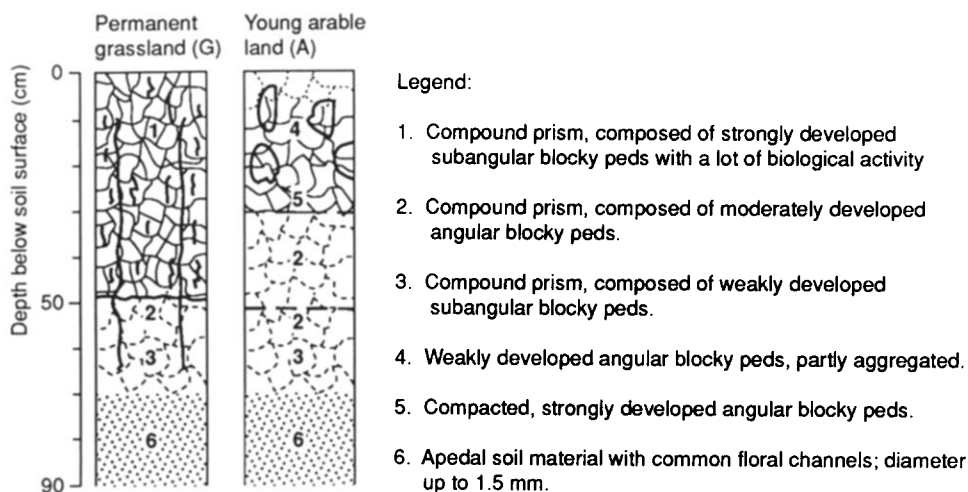


Fig. 1. Schematic description of two soil structure types, G and A, developed in a clay loam soil.

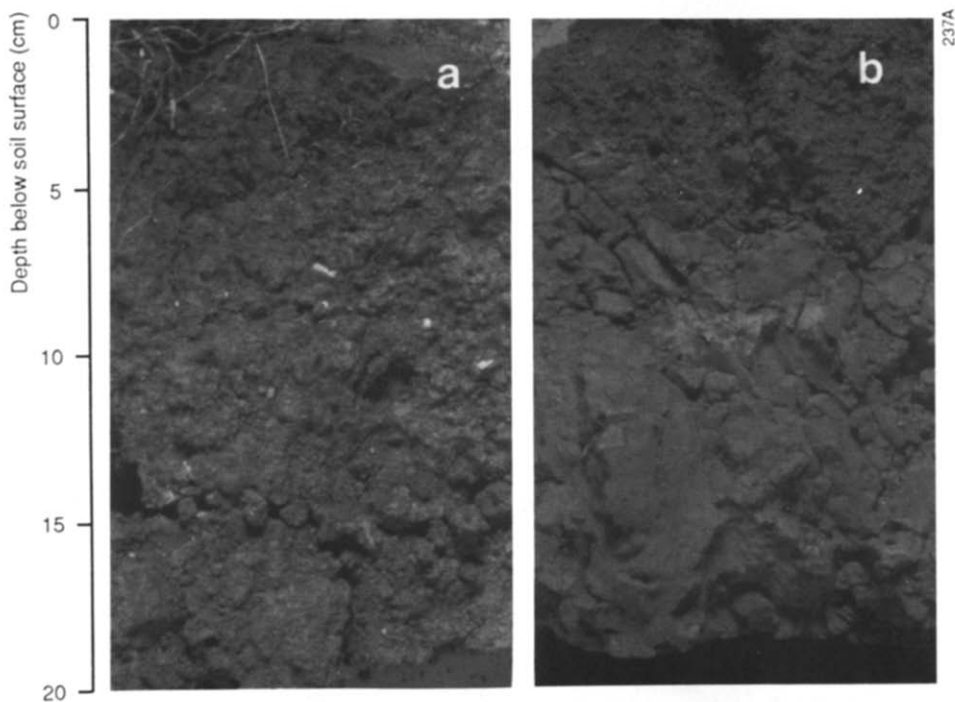


Fig. 2. The upper 20 cm of two soil structure types occurring in a clay loam soil: (a) permanent grassland; (b) young arable land.

the situation 8 years after the change from permanent grassland to land for market gardening. Water retention (θ - h) and hydraulic conductivity (K - h) curves were measured in the laboratory using conventional and new techniques (Bouma, 1983; Kooistra et al., 1984, 1985; Reinds, 1988b; Van Lanen and Boersma, 1988).

In clay loam soils, free water may move vertically downwards through macropores along an unsaturated soil matrix (bypass flow). The bypass flow was measured in undisturbed soil samples (height and diameter 20 cm) under a rainfall simulator (e.g. Van Stiphout et al., 1987). Because bypass flow is a function of the water content of the soil, the measurements were repeated for various field water conditions to derive relationships between daily rainfall, bypass flow and soil water content. Consequently, in May and June 1987 and in February 1988, both soil structure types were sampled (three samples each). Each of the soil structure types produced significantly different relationships, as reported by Van Lanen and Boersma (1988). The depth of infiltration of bypass flow into the soil matrix of the subsoil (internal catchment) had been measured in field experiments conducted on permanent grassland (Van Stiphout et al., 1987). Subsurface infiltration occurred mainly at the base of the cracks (transition of the clay loam topsoil into the sandy loam subsoil) at about a depth of 60 cm and at the base of worm channels at about a depth of 135 cm. Observations showed that worm channels in young arable land were predominantly discontinuous in the topsoil because of superficial tillage operations (Fig. 1). Therefore, bypass flow in young arable land was assumed to infiltrate only at the base of the cracks at about 60 cm depth.

Simulation model

Description

The soil-plant model (ONZAT) used had been proposed and validated by Van Drecht (1983, 1985). The model is based on principles of the SWATRE model (Belmans et al., 1983; Feddes et al., 1988b), which has been extensively validated for various crops, soils and climates (e.g. Van Wijk and Feddes, 1986; Hack-ten Broeke and Kabat, 1989). ONZAT simulates transient, one dimensional, soil-water flow in a vertical soil column which is subdivided in layers (thickness 5–15 cm). The model ONZAT has been extended with a submodel that describes bypass flow and internal catchment (Van Stiphout et al., 1987), and a procedure to consider effects of clods formed by soil management on crop water uptake (Bouma and Van Lanen, 1989).

The pressure heads, water contents and air-filled porosities as a function of depth and time, and the actual evaporation vs. time simulated by the simulation model ONZAT are of particular interest in the calculation of the land qualities. More information on the principles of the model, the boundary con-

ditions used and the simulation of the crop water uptake have been provided by Van Lanen et al. (1987).

Validation

The simulation model ONZAT was validated for the soils considered, using data measured in 1987. Grass and rhubarb were grown on the soils with a soil structure type of G and A, respectively. Measured groundwater depth on the experimental field and meteorological data from a nearby station were used

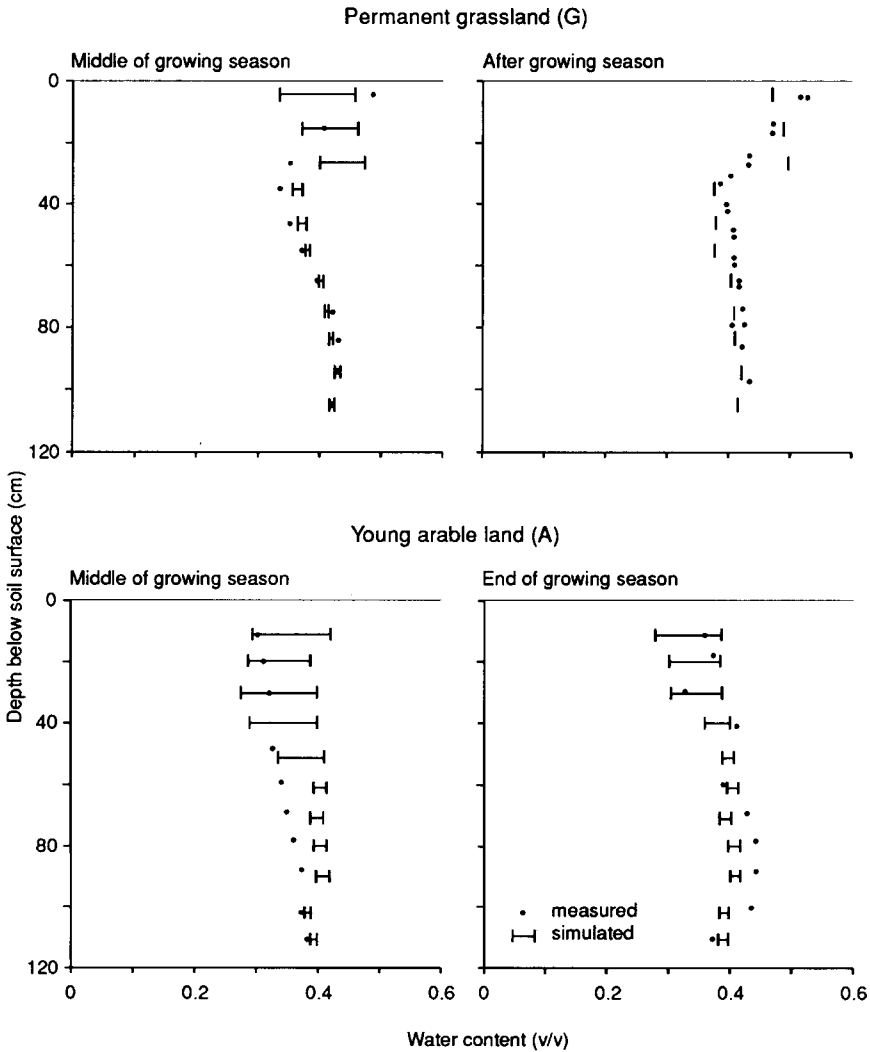


Fig. 3. Validation of the simulation model ONZAT comparing measured and simulated range of water contents in two soil structure types developed in a clay loam soil.

as boundary conditions. Potential evapotranspiration of both the grass and rhubarb crop were estimated using open-water evaporation and crop coefficients (e.g. Doorenbos and Kassam, 1979). The crop coefficients of the rhubarb crop consisted of the pronounced seasonal variation of crop height and soil cover and the early harvest. Bypass flow was derived from the measured relationships for both soil structure types as reported by Van Lanen and Boersma (1988). Rooting depth findings for permanent grassland were provided by Van Stiphout et al. (1987). The depth and distribution of rhubarb roots on young arable land were determined on a carefully prepared wall of a soil profile pit by using a transparent foil technique (e.g. Kooistra et al., 1985). Rooting patterns of the rhubarb crop and shapes and sizes of the soil peds were carefully analysed to ascertain whether the roots had poor accessibility to water (Bouma and Van Lanen, 1989).

Volumetric water contents at different depths and dates were measured to validate the model ONZAT, as illustrated in Fig. 3. Because of the variability in measured hydraulic properties and uncertainties, such as crop transpiration coefficients and harvest date (Reinds, 1988a), a range in simulated water contents had to be distinguished. The range was obtained by running ONZAT several times. A set of parameters was chosen for each run to fit the physical limits relevant to the considered experiments. The modification of the parameters hardly affected the simulated moisture contents for permanent grassland after the growing season. Thus, no range has been presented for this date. The simulated water contents corresponded reasonably well with the measured data. The coefficient of variance (CV) was used as a quantitative measure for the deviation between measured water contents and simulated water contents. The CV between measured and simulated water contents varied from 0.05 to 0.16. The CV was mostly between 0.08 and 0.1 (Reinds, 1988a).

Application of the model to derive land qualities

After satisfactory validation, the land qualities were simulated for both soil structure types. The simulations were made for the period 1955–1984 using meteorological data from a station operating in the southwest of the Netherlands. It thus enabled probability estimates to be provided for the various land qualities of both soil structure types, taking into account the temporal variability caused by the prevailing weather conditions. Potatoes were assumed to grow on both soil structure types because of their relative importance in terms of net revenues for the Dutch arable farmer. This somewhat hypothetical situation for the soil structure type G was used to illustrate the possible effects of soil structure differences. Potential evapotranspiration, water extraction by the potato roots and water table depth were approximated as described by Van Lanen et al. (1987). Rooting depths in both soil structure types were taken to be equal to those of unrestricted potato root growth in

sandy loam soils (e.g. Boone et al., 1985). Potatoes grown on both soil structure types were assumed to have a dense and homogeneous rooting pattern as demonstrated by the rhubarb crop.

Land qualities

The land qualities, water deficit, workability and soil aeration for both soil structure types in clay loam soils were determined according to the proposal by Van Lanen et al. (1987).

Water deficit

Water deficit over a certain period of time (e.g. total growing season) was defined as the cumulative difference between the potential and actual evapotranspiration, as calculated on a daily basis by ONZAT.

Workability

Workability was derived from a soil- and crop-specific threshold value (workability limit) and the simulated pressure heads at a depth of 5 cm. If the pressure head (h) in the topsoil of a clay loam soil is below a threshold value of $h = -70$ cm, moderate to good field conditions for potato planting should prevail (Van Wijk and Feddes, 1986). However, on days with h higher than the threshold value, planting cannot be accomplished without damaging the soil structure. A Boolean approach was followed, resulting in an adequate or an inadequate workability for each day. The probability of occurrence of a workable day in a certain period was determined by analysis of the simulated number of workable days in that particular period during the years 1955–1984.

Soil aeration

Simulated air-filled porosity in the topsoil was taken as the indicator for soil aeration. According to Bakker et al. (1987), if the air-filled porosity in clay loam soil was below a threshold value of $0.09 \text{ m}^3 \text{ m}^{-3}$, soil aeration was considered to be insufficient. The occurrence of days with adequate soil aeration was determined by comparing the simulated air-filled porosities with the soil-specific threshold value. Probability occurrence of days with sufficient aeration was calculated using a similar Boolean approach as done for workability.

Eventually, simple crop growth models were used to estimate the effect of the land qualities on crop yield. Therefore, differences in soil structure types owing to soil management systems were converted to impact on crop yield, providing a further quantification of soil structure degradation.

RESULTS

Soil structure and soil physical properties

Intensive tillage and traffic operations during 8 years caused significant differences in the volume fractions of different pores. Both pore shape and size had been affected, as shown in Fig. 4. Moreover, in the lower part of the top-soil (10–30 cm) total macro-porosity (volume fractions of different pores larger than 30 μm) was substantially higher in the soil structure type G than in type A. Furthermore, the volume of round pores, representing worm activity, was considerably greater in type G.

Soil structure observations showed that because it was easy for the compound prisms or aggregates (Fig. 1) to be broken down to relatively small peds, no high mechanic resistances for root growth occurred. This was sup-

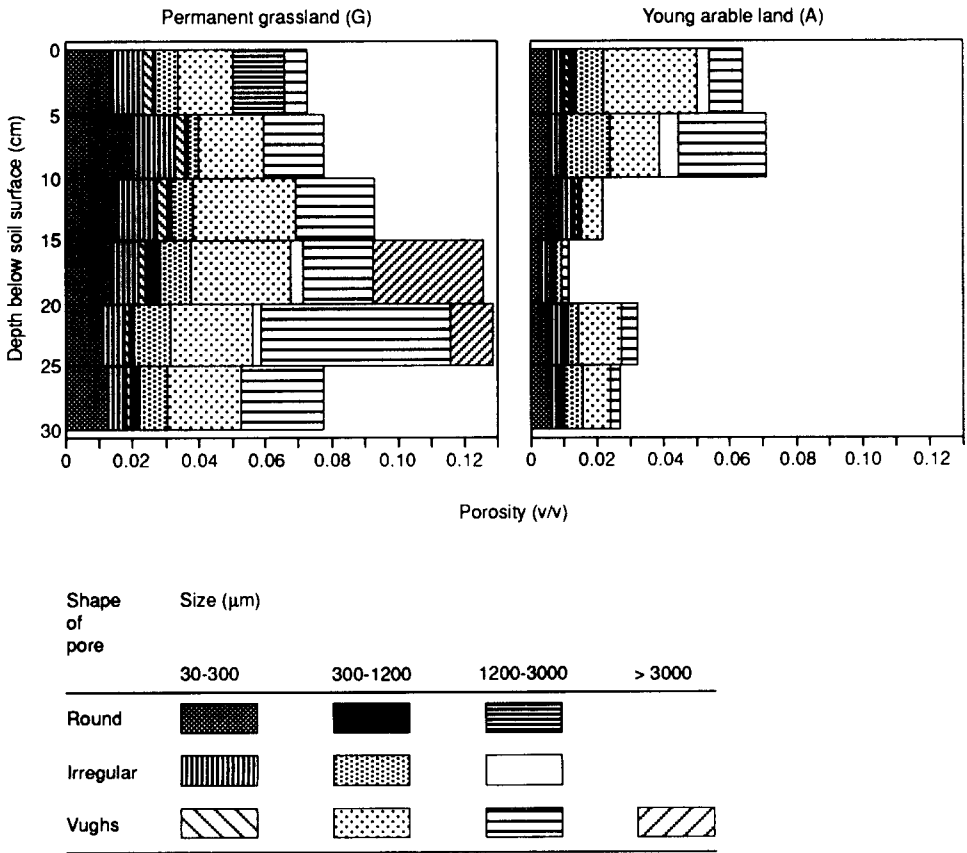


Fig. 4. Volume fraction of pores (greater than 30 μm) distributed over different pore-shape and three pore-size classes for two soil structure types in a clay loam soil.

ported by the well distributed roots of the rhubarb crop observed. The dense and homogeneous rooting pattern and unsaturated hydraulic conductivity of these clay loam soils resulted in relatively high volumes of readily accessible soil water in the peds. Therefore, it was not necessary to adapt the sink term as proposed by Bouma and Van Lanen (1989), irrespective of the soil structure type considered in our study.

Representative curves for soil water retention and hydraulic conductivity for the topsoil of both soil structure types were derived from individual samples (Fig. 5). The range of soil water retention data of these individual samples at different depths has been presented by Van Lanen and Boersma (1988). Water contents in the topsoil of type G were up to $\log(-h) = 3.0$, significantly higher than in type A. Above $\log(-h) = 3.0$ the water contents of type A were slightly higher than those of type G. These differences in water contents indicated a higher available soil water in type G. The unsaturated hydraulic conductivities of the topsoil of type G were clearly higher than the ones of type A with the exception of the range of pressure heads close to saturation. Generally, these conductivity differences offer better conditions for up- and downwards flow of soil water in type G.

After 8 years of different soil management, only slight changes occurred in the subsoil revealing minor differences between physical data of the subsoils of the two soil structure types (Table 1; Reinds, 1988b). At a depth of 40–50 cm, the total macro-porosity of type A was somewhat (more than 1%) less than the porosity of type G, probably indicating the beginning of subsoil compaction. However, differences in macro-porosity of less than 1% at a depth of 30–40 cm did not clearly support this supposition.

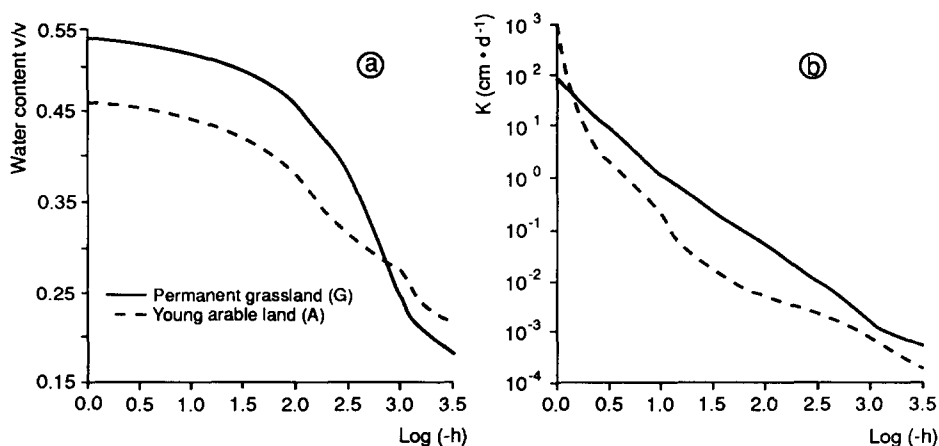


Fig. 5. Representative soil-water retention (a) and hydraulic conductivity curve (b) of the upper 25 cm of two soil structure types developed in a clay loam soil.

TABLE 1

Total macro-porosity of pores greater than $30\ \mu\text{m}$ in the subsoil of the two soil structure types in identical clay loam soils

Depth below soil surface (cm)	Porosity as volume fraction	
	Permanent grassland	Young arable land
30–40	0.080	0.085
40–50	0.092	0.079
50–60	0.095	0.086

Land qualities

Water deficit

The susceptibility to drought of these clay loam soils is low under Dutch climatic conditions, irrespective of the soil structure type. The simulated mean annual water deficit of potatoes grown on soils of a soil structure type A was 36 mm, whereas on soils of type G the deficit was slightly lower, i.e. 32 mm. The small water deficit was unequally distributed throughout the growing season. For both soil structure types differences between the potential and the actual evapotranspiration were mainly restricted to the period 20 May–20 June.

The somewhat higher susceptibility to drought of type A compared with G is expressed in Fig. 6. In 90 out of 100 years the water deficit of potatoes

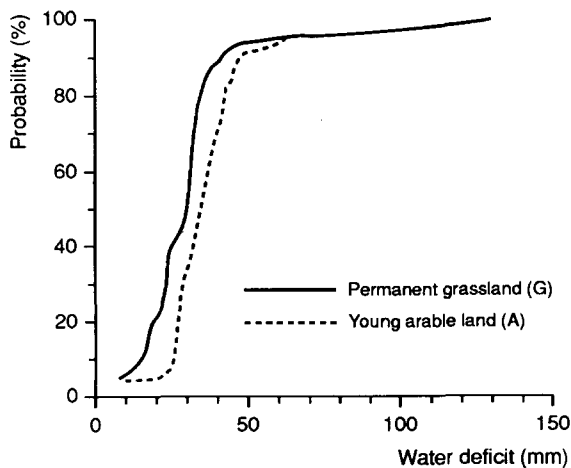


Fig. 6. Cumulative probability occurrence of annual water deficits of potatoes grown on two different soil structure types of a clay loam soil.

grown on soils with type A is predicted to be less than 50 mm, while on soils with type G the deficit is predicted to be less than 40 mm.

The effect of water stress on yield is dependent on the phenological stage of the potato crop (e.g. Van Loon, 1981; Roth and Roth, 1985). In the growth stage after flowering, stress will certainly reduce tuber yield. In an earlier stage the effect depends on several aspects, such as the leaf area index, subsequent respiration losses and the possible change in the partitioning of the photosynthesis products. A small amount of water stress at an early stage of growth may even be beneficial for the final yield of tubers. According to a simple crop growth model proposed by the Werkgroep HELP (1987) a mean water deficit difference of 4 mm between both soil structure types in these soils approximately equalled about 1.8% tuber yield difference.

Workability

The probability of a workable day for any day of the year relevant to field machinery work is indicated in Fig. 7. In the Netherlands according to Alblas et al. (1987), the important periods for machinery work for potato growing are 15 March–30 April (planting of potatoes) and 15 June–15 October (harvesting of potatoes). Figure 7 was derived from simulated 24 h data over a period of 30 years. The median probability of having a workable day on a soil with soil structure type A was lower than that with type G (Fig. 7(a)). For example, for half of the period the probability of occurrence of a workable day in mid-April would be less than 20% for soils with type A, whereas on soils with type G this probability would be about 50%. In September and October, the differences would be even more pronounced.

The range in probability (10–90% prediction range) of occurrence of a

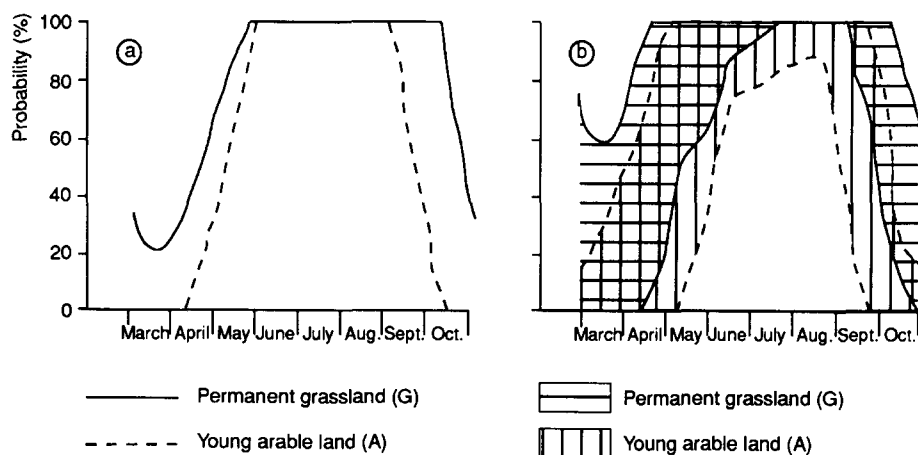


Fig. 7. Probability of occurrence of a workable day on two different soil structure types of a clay loam soil: (a) median; (b) 10–90% prediction range.

workable day provides information on workability in 80% of the years (Fig. 7(b)). This range was derived from the earlier mentioned record of 30-year meteorological data. The probability ranges of both soil structure types partly coincide (shaded area represented by squares). Sections of the figure with only vertical lines imply that in those periods, in part of the period considered, the workability of soils with a soil structure type A was worse than that of type G soils. Conversely, only horizontal lines indicate a better workability in part of the considered years of soils with type G. For example, in mid-April, in 80% of the years, the probability of occurrence of a workable day was between 0 and 70% on soils with a type A soil structure. For soils with a type G soil structure the limits of the 80% probability range were higher in the same period, i.e. 5 and 95%. In September and October, the differences were more pronounced, as already indicated by the median. For example, in mid-September, in 80% of the years it was possible for the probability of having a workable day to vary from 40 to 100% on soils with a type A soil structure, while on type G soils hardly any days occurred with inadequate workability.

A higher probability of occurrence of workable days results in the possibility of a longer growing season. Generally, a yield reduction of about 0.1–0.2% day⁻¹ in April and 0.3–1.3% day⁻¹ in May occurred when potatoes were planted after the optimal date (e.g. Whitney and Elbanna, 1985). Workability restrictions were assessed and seen to cause a potato yield depression of 2% or more in 50% of the years on type G soils and in 70% of the years on type A soils. Yield differences of 5% or more were estimated to occur in 20 out of 100 years between both soil structure types and differences of 10% or more in 10 out of 100 years.

Soil aeration

Soil aeration was derived from simulated 24 h data of the air-filled porosity over a period covering 30 years. Because air-filled porosity and workability are linked through the simulated pressure head and the soil water retention characteristic, similar graphs to Fig. 7 can be produced for soil aeration (Van Lanen et al., 1987; Reinds, 1988a). Therefore, in Fig. 8 the air-filled porosities for both soil structure types are presented in a somewhat different manner. For example, in the second 10-day period of April, in about 50% of the years the air-filled porosity in soils with a type A soil structure is predicted to be less than 0.05 m³ m⁻³, indicating unfavourable conditions for crop growth. In type G soils this percentage equalled about 25%. In the initial part of the growing season, the threshold value of 0.09 m³ m⁻³, was exceeded only in about 10% of the years, irrespective of the soil structure type. In the final stage of the growing season (1–10 September), the clay loam soils were better aerated than at the start. In soils with type A soil structure, an inadequate soil aeration (air-filled porosity of less than 0.09 m³ m⁻³) only occurred in about

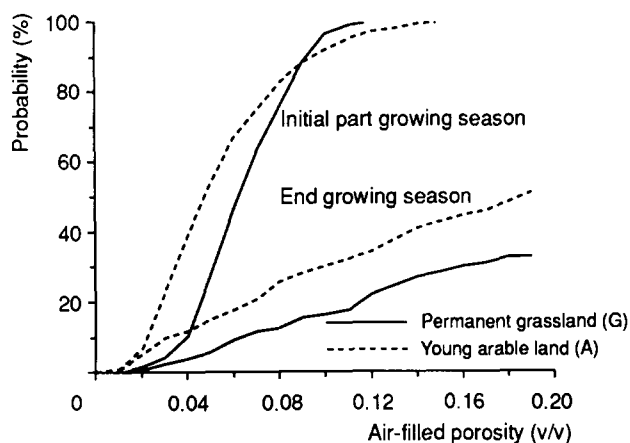


Fig. 8. Cumulative probability occurrence of air-filled porosities for two 10-day periods (initial and final part of growing season) of two different soil structure types developed in a clay loam soil.

30% of the years. The soils with type G soil structure were better aerated. An insufficient aeration only occurred in about 15% of the years. Therefore, in the first 10-day period of September, there was a difference of 15%. The difference between both soil structure types in the percentage of years with an inadequate soil aeration are somewhat less in the major part of the growing season, e.g. 5–10% in May–August (Reinds, 1988a).

The impact on the potato yield of soil aeration is difficult to assess because of direct effects (dry matter increase) and indirect effects (e.g. phytosanitary problems). The estimated direct effect on these well-drained soils under average Dutch climatic conditions according to simple crop growth models is probably about a 2% difference in tuber yield between the soil structure types. In about 50 out of 100 years the yield difference was assessed to be 2.5% or more between the soil structure types and in 10 out of 100 years, more than 5%.

CONCLUSIONS AND DISCUSSION

Intensive tillage and traffic operations on clay loam soils used for field vegetable growing over a period of 8 years resulted in significantly different soil structure types compared with otherwise identical soils used as permanent grassland. Different soil structural types could clearly be recognised in the field and subsequently characterised in terms of macro- and micromorphological and soil hydraulic properties. Associated calculated differences in water deficits, soil aeration and workability prove the functional quality of the permanent grassland to be higher than that of the young arable land. A higher

potato tuber yield of at least 5% in 20 out of 100 years would be expected on Dutch clay loam soils with the favourable permanent grassland soil structure compared with the less favourable young arable land structure. These yield differences between the same soils with different soil structures seem to be realistic given that similar or even more pronounced measured differences were reported by Feddes et al. (1988a) and Van Loon and Bouma (1988).

A prolonged application of the intensive soil management system on young arable land could be expected to result in a compacted subsoil, i.e. the formation of ploughpans. Under Dutch climatic conditions a disruption of ploughpans usually results in more serious soil compaction after a few years (secondary ploughpans), as illustrated by Kooistra et al. (1984, 1985) and Van Lanen et al. (1987).

The results of our study can also be used to explore the maximum possible effect on soil structure regeneration by for example, increasing the organic matter content, reducing root crops in the rotation and by implementing technical alternatives for current traffic and tillage systems, as suggested by Håkansson et al. (1988). If we assume that a soil structure type of permanent grassland, as recognised in our study, is the most favourable soil structure, differences in water deficits, workability and soil aeration between a particular soil structure and that of permanent grassland can be interpreted in terms of what farmers can theoretically attain. So, if in our case of clay loam soils, the soil structure of young arable land could be changed to the more favourable one of permanent grassland, then the maximum benefit to the farmer would be a 10% reduction in water deficits, 40% more machinery work days in April and about 5–to 10% more days with an adequately aerated soil within the growing season.

Although comprehensive, integrated machine–soil–crop models still have serious shortcomings (e.g. Hadas et al., 1988), the results of our study show that submodels, such as the soil–crop model supplied with parameterised soil structure types could be successfully applied to quantify negative effects of soil degradation and to explore possible positive effects of soil structure regeneration. Moreover, the results could be used for a further development of the integrated model. Morphological characterisation of soil structure forms an integrated part of the procedure.

Simulation modelling combined with field experiments which have only been conducted for a few years enables us to extrapolate results to a wider range of site conditions. Effects of soil degradation may be transferred to a wide range of weather conditions by using a historical record of weather data, as shown in our study. Monitoring these effects on experimental fields over a long period to obtain similar temporal variability expressions, is nearly impossible because of budgetary restrictions. The introduction of simulation modelling in short-term conventional agronomic tillage experiments in order to obtain a better understanding would be very attractive. This would allow

us more opportunities to transfer knowledge to other agro-ecological conditions. Then, however, a prerequisite would be the collection of more soil physical data in addition to those such as penetrometer resistances, bulk densities, pore volumes and air-filled porosity at $\log(-h) = 2.0$.

On a farm scale, soil structure types can reproducibly be recognised in the field by soil surveyors using macro-morphological techniques. Compared with conventional surveys, recognition of soil structure phases on large-scale soil maps is invaluable to reduce variability within mapping units. Description and classification of soil structure do not need to be elaborated any further for this purpose, efforts should be concentrated on a further characterisation of significantly different types by applying simple physical field tests, such as measurement of cone resistances, functional monitoring of soil water contents, pressure heads, gas concentrations, rooting patterns and analysis of infiltration experiments, possibly with tracers. Additionally, micro-morphological and soil physical analyses need to be made at the laboratory, as demonstrated in our study, to enable more realistic boundary conditions (e.g. soil structure type dependent bypass and internal catchment) to be defined and to allow a more precise determination of parameters governing the flow processes in the soil. This knowledge is indispensable if realistic answers on current land use problems, which include soil management, crop production and environmental topics are to be provided.

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